

Govind Mudavadkar

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EDUCATION

Northeastern University, Boston, MA | *M.S. Computer Engineering* | GPA: 3.8/4.0

Sep 2023 – Dec 2025

University of Pune, India | *B.E. Computer Science* | GPA: 3.8/4.0

Jun 2019 – May 2023

EXPERIENCE

AI/ML Platform Engineer | Pfizer Inc, Groton, CT

Jan 2026 – Present

- Architected **agentic AI platform** for enterprise document intelligence and real-time question answering, on **AWS** saving **800+** hours annually
- Engineered multimodal AI pipelines retrieval on **AWS(Bedrock, S3, Lambda)** enabling faster real time inference, reducing latency from **2 minutes to 50 seconds** through retrieval tuning, prompt/token optimization (LLMLingua compression + semantic caching), and model orchestration
- Designed conversational agent workflows in the **LangGraph** (Langchain ecosystem) with reusable nodes for multi-step reasoning, integrating the Model Context Protocol (**MCP**) to build automated document generation tools that eliminated **25%** of manual operational overhead
- Built and deployed an **AI assistant platform** using **Python, FastAPI, React (TypeScript), and SQL** on **AWS, Dataiku**, containerized with **Docker and Kubernetes** for scalable, real-time inference serving 20+ scientists
- Built **LLM/agent evaluation framework** with LangSmith and **RAGAS** measuring faithfulness, answer relevance, and hallucination rate across an eval set of **300+** queries, cutting hallucination rate to under **5%**
- Implemented production **guardrails** (PII redaction, prompt-injection filtering) via **Bedrock**, and routed traffic through an **LLM gateway** (LiteLLM) for model routing and rate limiting cutting API costs **25%**

Software Development Automation Engineer Co-Op | Pfizer Inc, Groton, CT

Jan 2025 – Sep 2025

- Architected an in-house multimodal AI platform combining **LLMs, RAG, vector databases**, and knowledge graphs, supporting **400+** daily queries across automated document intelligence and data exploration workflows
- Reduced **MySQL/PostgreSQL** query execution time by **18%** by redesigning indexes, query optimization and refactoring SQL for critical workloads
- Built a **React.js** application for predictive ML model evaluation, improving validation accuracy and reducing decision-making cost, time by **60%**
- Engineered scalable **REST APIs and microservices** using **Java, Spring Boot, Apache Camel, and Docker**, supporting **200K+ daily** data claim requests per day at p95 response time under **2-second**

Software Development, Research Assistant | Northeastern University, Boston, MA

Feb 2024 – Dec 2024

- Developed custom **C++** integration wrappers using LibTorch to bridge PyTorch models with laboratory data logging infrastructure, automating diagnostic workflow while eliminating **\$10K** in hardware upgrade costs
- Built an end-to-end distributed image processing pipeline on **AWS S3 and SageMaker**, automating storage, retrieval, and database integration to process **500K+** images via multithreading, improving diagnostic accuracy by **81%**
- Exported **ML models** to ONNX, ran INT8 inference to batch-process **500K+** images, improving throughput by **2.5X** without diagnostic accuracy loss
- Implemented PageRank algorithm using **Spark**, integrating results with Elasticsearch to enhance search relevance score

Software Development Engineer, Intern | BMC Software Inc, Pune, India

Jan 2022 – Jul 2022

- Developed a custom VS Code extension using Vue.js, WebView API and **Spring Boot** backend to automate **microservices** metadata retrieval, reducing manual data entry by **63%** and eliminating context switching between IDEs and databases
- Assembled a robust backend using Maven, **PostgreSQL, REST APIs, and MVC** architecture to support scalable application functionality
- Followed **Agile and SDLC** best practices, using GitHub for version control, JIRA for task management, and **JUnit** for testing automation

SKILLS

Programming & Scripting: Python, Java, TypeScript, JavaScript, React, C++, C#, SQL, Bash/Linux

Backend & Frameworks: Spring Boot, MVC, FastAPI, REST APIs, Microservices, Next.js, Angular, Tailwind

AI & Machine Learning: Generative AI (LLMs, RAG), Agentic AI (LangGraph), MCP (Model Context Protocol), Computer Vision, NLP, Statistics

Data Systems & Orchestration: PostgreSQL, MySQL, NoSQL, MongoDB, Redis Elasticsearch, Apache (Kafka, Camel), MapReduce/Spark

Cloud & Infrastructure: AWS (Bedrock, S3, EC2, Lambda, , DynamoDB), Microsoft Azure (AI Foundry, Document Intelligence), GCP

DevOps & Tools: Kubernetes (K8s), Docker, CI/CD Pipelines (Jenkins) Git/GitHub, Agile/Scrum Methodology, JUnit, Object-Oriented Programming (OOP), Data Structures, MLOps

ACADEMIC PROJECTS

Automated Cloud Deployment Pipeline | AWS, GitHub Actions, Infrastructure-as-Code

- Architected cloud native user onboarding system on **AWS** using **SNS, Lambda, SendGrid**, with secure VPC boundaries, security groups, ingress rules
- Automated CI/CD with **EC2 Auto Scaling, ELB, GitHub Actions** (tests,AMIs, rolling updates), reducing downtime by **35%**

Talent Palette Web Application | Web Development, Cloud Development, OOP

- Crafted responsive MERN platform for artists with **React, TypeScript**, and OOP, enabling job creation & content feeds
- Developed scalable **REST APIs with JWT auth**, improved content discovery by **20%**, and deployed on AWS EC2 with Kubernetes and Jenkins

PUBLICATIONS

- KnovaAI: Advanced Multimodal AI for Chemistry- Aware Document Intelligence** | *AICHe 2026* | [Link](#)
- Additional 4+ peer-reviewed research available via Google Scholar link** | [Google Scholar](#) | [Link](#)